

US00PP34659P2

(12) **United States Plant Patent**  
**Nguyen et al.**

(10) **Patent No.:** **US PP34,659 P2**  
(45) **Date of Patent:** **Oct. 18, 2022**

- (54) **STRAWBERRY PLANT NAMED**  
**‘DRISSTRAWNINETYFOUR’**
- (50) Latin Name: *Fragaria x ananassa*  
Varietal Denomination: **DrisStrawNinetyFour**
- (71) Applicant: **Driscoll’s, Inc.**, Watsonville, CA (US)
- (72) Inventors: **Phuong Nguyen**, Watsonville, CA  
(US); **Michael D. Ferguson**,  
Watsonville, CA (US)
- (73) Assignee: **Driscoll’s, Inc.**, Watsonville, CA (US)
- (\*) Notice: Subject to any disclaimer, the term of this  
patent is extended or adjusted under 35  
U.S.C. 154(b) by 0 days.
- (21) Appl. No.: **17/581,752**
- (22) Filed: **Jan. 21, 2022**
- (51) **Int. Cl.**  
*A01H 5/08* (2018.01)  
*A01H 6/74* (2018.01)
- (52) **U.S. Cl.**  
USPC ..... **Plt./209**  
CPC ..... *A01H 6/7409* (2018.05)
- (58) **Field of Classification Search**  
USPC ..... Plt./209  
CPC ..... *A01H 6/7409*  
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

|            |         |                     |
|------------|---------|---------------------|
| PP1,745 P  | 8/1958  | Lang                |
| PP3,981 P  | 11/1976 | Bringhurst et al.   |
| PP4,487 P  | 11/1979 | Bringhurst et al.   |
| PP4,538 P  | 5/1980  | Bringhurst et al.   |
| PP5,262 P  | 7/1984  | Voth et al.         |
| PP5,265 P  | 7/1984  | Voth et al.         |
| PP5,266 P  | 7/1984  | Bringhurst et al.   |
| PP5,300 P  | 10/1984 | Johnson, Jr. et al. |
| PP5,840 P  | 12/1986 | Johnson, Jr. et al. |
| PP6,191 P  | 5/1988  | Johnson, Jr. et al. |
| PP6,231 P  | 7/1988  | Johnson, Jr. et al. |
| PP6,578 P  | 1/1989  | Voth et al.         |
| PP6,579 P  | 1/1989  | Bringhurst et al.   |
| PP7,024 P  | 9/1989  | Johnson, Jr. et al. |
| PP7,172 P  | 2/1990  | Voth et al.         |
| PP7,522 P  | 5/1991  | Johnson, Jr. et al. |
| PP7,614 P  | 8/1991  | Bringhurst et al.   |
| PP7,615 P  | 8/1991  | Bringhurst et al.   |
| PP8,086 P  | 1/1993  | Nelson et al.       |
| PP8,205 P  | 4/1993  | Nelson et al.       |
| PP8,649 P  | 3/1994  | Sjulin et al.       |
| PP8,661 P  | 3/1994  | Bringhurst et al.   |
| PP8,708 P  | 5/1994  | Voth et al.         |
| PP8,745 P  | 5/1994  | Sjulin et al.       |
| PP9,130 P  | 5/1995  | Sjulin et al.       |
| PP9,909 P  | 6/1997  | Ackerman et al.     |
| PP10,221 P | 2/1998  | Sjulin et al.       |
| PP10,534 P | 8/1998  | Sjulin et al.       |
| PP10,642 P | 10/1998 | Amorao et al.       |
| PP11,035 P | 8/1999  | Mowrey et al.       |
| PP11,277 P | 3/2000  | Gilford et al.      |
| PP11,279 P | 3/2000  | Gilford et al.      |
| PP11,522 P | 9/2000  | Amorao et al.       |

|             |         |                 |
|-------------|---------|-----------------|
| PP11,548 P  | 10/2000 | Amorao et al.   |
| PP11,554 P  | 10/2000 | Sjulin et al.   |
| PP11,639 P  | 11/2000 | Mowrey et al.   |
| PP12,186 P2 | 11/2001 | Gilford et al.  |
| PP12,436 P2 | 3/2002  | Amorao et al.   |
| PP12,577 P2 | 4/2002  | Amorao et al.   |
| PP12,817 P2 | 7/2002  | Gilford et al.  |
| PP12,899 P2 | 9/2002  | Mowrey et al.   |
| PP13,386 P2 | 12/2002 | Mowrey et al.   |
| PP13,469 P3 | 1/2003  | Larson et al.   |
| PP14,005 P3 | 7/2003  | Amorao et al.   |
| PP14,062 P3 | 8/2003  | Amorao et al.   |
| PP14,109 P3 | 8/2003  | Gilford et al.  |
| PP14,771 P3 | 5/2004  | Amorao et al.   |
| PP15,145 P2 | 9/2004  | Mowrey et al.   |
| PP15,308 P2 | 11/2004 | Sjulin et al.   |
| PP15,375 P2 | 11/2004 | Mowrey et al.   |
| PP15,435 P2 | 12/2004 | Sjulin et al.   |
| PP15,596 P2 | 3/2005  | Amorao et al.   |
| PP15,731 P2 | 4/2005  | Amorao et al.   |
| PP15,752 P2 | 5/2005  | Gilford et al.  |
| PP16,070 P2 | 10/2005 | Gilford et al.  |
| PP16,238 P2 | 2/2006  | Amorao et al.   |
| PP16,241 P2 | 2/2006  | Mowrey et al.   |
| PP16,298 P2 | 2/2006  | Gilford et al.  |
| PP16,299 P2 | 2/2006  | Gilford et al.  |
| PP16,475 P2 | 4/2006  | Gilford et al.  |
| PP16,558 P3 | 5/2006  | Lopez           |
| PP18,000 P2 | 9/2007  | Meulenbroek     |
| PP18,040 P3 | 9/2007  | Mowrey et al.   |
| PP18,041 P3 | 9/2007  | Gilford         |
| PP18,458 P2 | 1/2008  | Ferguson et al. |
| PP18,575 P3 | 3/2008  | Amorao et al.   |
| PP18,878 P2 | 6/2008  | Mowrey et al.   |
| PP19,240 P2 | 9/2008  | Gilford et al.  |
| PP19,673 P3 | 2/2009  | Ferguson et al. |
| PP19,767 P2 | 2/2009  | Shaw et al.     |
| PP20,248 P3 | 9/2009  | Rogers et al.   |
| PP20,363 P2 | 9/2009  | Chandler        |
| PP20,701 P2 | 2/2010  | Gilford et al.  |
| PP20,731 P2 | 2/2010  | Mowrey et al.   |
| PP20,733 P2 | 2/2010  | Mowrey et al.   |
| PP20,735 P2 | 2/2010  | Ferguson        |
| PP20,775 P2 | 2/2010  | Mowrey et al.   |
| PP20,922 P2 | 4/2010  | Gilford et al.  |
| PP21,538 P2 | 11/2010 | Gilford et al.  |
| PP21,559 P2 | 12/2010 | Ferguson et al. |
| PP21,762 P2 | 3/2011  | Stewart et al.  |
| PP22,040 P3 | 7/2011  | Stewart et al.  |
| PP22,218 P2 | 11/2011 | Ferguson et al. |
| PP22,247 P2 | 11/2011 | Ferguson        |
| PP23,107 P2 | 10/2012 | Ferguson et al. |
| PP23,148 P2 | 10/2012 | Gilford et al.  |
| PP23,377 P2 | 2/2013  | Ferguson et al. |

(Continued)

Primary Examiner — Anne Marie Grunberg

(74) *Attorney, Agent, or Firm* — Morrison & Foerster  
LLP

(57) **ABSTRACT**

A new and distinct variety of strawberry plant named ‘DrisStrawNinetyFour’, particularly selected for its large berry size, high vigor, long trusses, and flavor of the fruit, is disclosed.



(56)

References Cited

U.S. PATENT DOCUMENTS

PP23,378 P2 2/2013 Pullen et al.  
PP23,382 P2 2/2013 Ferguson et al.  
PP23,383 P2 2/2013 Ferguson et al.  
PP23,400 P2 2/2013 Ferguson et al.  
PP23,401 P2 2/2013 Pullen et al.  
PP23,459 P2 3/2013 Stewart et al.  
PP23,506 P3 4/2013 Ferguson et al.  
PP23,517 P3 4/2013 Ferguson et al.  
PP24,096 P3 12/2013 Fear et al.  
PP24,317 P3 3/2014 Ferguson et al.  
PP24,333 P3 3/2014 Vitten et al.  
PP24,395 P3 4/2014 Vitten et al.  
PP24,533 P3 6/2014 Ferguson et al.  
PP24,745 P2 8/2014 Vitten et al.  
PP25,408 P3 4/2015 Vitten et al.  
PP25,437 P3 4/2015 Vitten et al.  
PP25,698 P3 7/2015 Ferguson et al.  
PP25,699 P3 7/2015 Stewart et al.  
PP25,747 P3 7/2015 Kibbe et al.  
PP25,866 P3 9/2015 Ferguson et al.  
PP26,800 P3 6/2016 Stewart et al.  
PP26,801 P3 6/2016 Stewart et al.  
PP26,802 P3 6/2016 Rodriguez Alcazar et al.  
PP27,442 P2 12/2016 Kibbe et al.  
PP27,645 P3 2/2017 Vitten et al.  
PP27,682 P3 2/2017 Kibbe et al.  
PP27,711 P3 2/2017 Vitten et al.  
PP27,813 P3 3/2017 Ferguson et al.  
PP29,289 P3 5/2018 Vitten et al.  
PP29,728 P2 10/2018 Stewart et al.  
PP29,729 P2 10/2018 Kibbe et al.  
PP29,730 P2 10/2018 Kibbe et al.

PP29,731 P2 10/2018 Ferguson et al.  
PP29,747 P2 10/2018 Vitten et al.  
PP29,748 P2 10/2018 Vitten et al.  
PP29,749 P2 10/2018 Stewart et al.  
PP30,775 P2 8/2019 Carrillo Mendoza et al.  
PP30,789 P2 8/2019 Ferguson et al.  
PP30,818 P2 8/2019 Pakozdi et al.  
PP30,851 P2 8/2019 Fear et al.  
PP30,936 P2 10/2019 Stewart et al.  
PP31,083 P2 11/2019 Carrillo Mendoza et al.  
PP31,233 P2 12/2019 Pakozdi et al.  
PP31,527 P3 3/2020 Carrillo Mendoza et al.  
PP31,655 P2 4/2020 Carrillo Mendoza et al.  
PP31,703 P2 4/2020 Stewart et al.  
PP31,827 P2 6/2020 Ferguson et al.  
PP31,896 P2 6/2020 Pakozdi et al.  
PP31,935 P2 7/2020 Pakozdi et al.  
PP32,079 P2 8/2020 Carrillo Mendoza et al.  
PP32,080 P2 8/2020 Ferguson et al.  
PP32,271 P2 10/2020 Ferguson et al.  
PP32,305 P3 10/2020 Pakozdi et al.  
PP32,498 P2 11/2020 Mendoza et al.  
PP32,499 P3 11/2020 Jacobs et al.  
PP32,500 P3 11/2020 Stewart et al.  
PP32,801 P2 2/2021 Ferguson et al.  
PP32,814 P2 2/2021 Stewart et al.  
PP32,824 P2 2/2021 Stewart et al.  
PP33,070 P2 5/2021 Mendoza et al.  
PP33,090 P2 5/2021 Mendoza et al.  
PP33,283 P2 7/2021 Stewart et al.  
PP33,513 P2 9/2021 Stewart et al.  
PP33,737 P2 12/2021 Hernandez et al.  
PP33,738 P2 12/2021 Stewart et al.  
2003/0079263 P1 4/2003 Gilford et al.  
2013/0276182 P1 10/2013 Fear et al.



## STRAWBERRY PLANT NAMED 'DRISSTRAWNINETYFOUR'

Latin name:

Botanical classification: *Fragaria* x *ananassa*.

Varietal Denomination: The varietal denomination of the claimed variety of strawberry plant is 'DrisStrawNinetyFour'.

### BACKGROUND OF THE INVENTION

Cultivated strawberry is a hybrid species of the genus *Fragaria* that is grown worldwide for its fruit. Modern strawberry was first bred in Brittany, France, in the 18<sup>th</sup> century by crossing *Fragaria virginiana* with *Fragaria chiloensis*. Strawberry fruit is an aggregate accessory fruit, with the fleshy part of the fruit being derived from the receptacle that holds the ovaries.

Strawberry varieties vary widely in color, size, shape, flavor, season of ripening, degree of fertility, and susceptibility to disease. Certain varieties vary in foliage, and some vary in the relative development of their reproductive organs. Typically, strawberry flowers appear hermaphroditic in structure, but function as either male or female. Generally, commercial production of strawberry plants involves propagation from runners and distribution as either plugs or bare root plants. Cultivation is either perennial or annual plasticulture. During the off season, strawberries can also be produced in greenhouses.

Strawberry fruit is widely appreciated for its characteristic bright red color, aroma, juicy texture, and sweetness. Strawberry fruit is a popular fruit that is generally consumed either fresh or in prepared foods, such as preserves and baked goods.

Strawberry is an important and valuable fruit crop. Accordingly, there is a need for new varieties of strawberry plants. In particular, there is a need for improved varieties of strawberry plant that are stable, high yielding, and agronomically sound.

### SUMMARY OF THE INVENTION

In order to meet these needs, the present invention is directed to an improved variety of strawberry plant. In particular, the invention relates to a new and distinct variety of strawberry plant (*Fragaria* x *ananassa*), which has been denominated as 'DrisStrawNinetyFour'.

Strawberry plant variety 'DrisStrawNinetyFour' originated from a controlled cross between the female parent 'DrisStrawSixtyNine' (U.S. Plant Pat. No. 32,801) and the proprietary male parent '575AC159' (unpatented). Progeny plants from this cross, including 'DrisStrawNinetyFour', were asexually propagated via stolons in Shasta County, Calif. in March 2015. Strawberry plant variety 'DrisStrawNinetyFour' was later specifically identified and selected in Ventura County, Calif. in October 2015.

'DrisStrawNinetyFour' was subsequently asexually propagated via stolons, and has undergone testing at test plots in Ventura County, Calif. for four years (2016 to 2020). The present variety has been found to be stable and reproduce true to type through successive asexual propagations via stolons and tissue culture.

'DrisStrawNinetyFour' was particularly selected for its large berry size, high vigor, long trusses, and flavor of the fruit.

### DESCRIPTION OF THE DRAWINGS

This new strawberry plant is illustrated by the accompanying photographs. The colors shown are as true as can be

reasonably obtained by conventional photographic procedures. Unless otherwise indicated, the photographs are of plants that are five months old.

FIG. 1 illustrates whole fruit of variety 'DrisStrawNinetyFour'.

FIG. 2 illustrates longitudinal sections of fruit of variety 'DrisStrawNinetyFour'.

FIG. 3 illustrates the upper surface (top row) and lower surface (bottom row) of flowers of variety 'DrisStrawNinetyFour'.

FIG. 4 illustrates the lower surface (top) and upper surface (bottom) of leaves of variety 'DrisStrawNinetyFour'.

FIG. 5 illustrates a whole plant of variety 'DrisStrawNinetyFour'.

### DETAILED BOTANICAL DESCRIPTION

The following detailed descriptions set forth the distinctive characteristics of 'DrisStrawNinetyFour'. The data which define these characteristics is based on observations taken in Ventura County, Calif. from 2016 to 2020. This description is in accordance with UPOV terminology. Color designations, color descriptions, and other phenotypical descriptions may deviate from the stated values and descriptions depending upon variation in environmental, seasonal, climatic, and cultural conditions. 'DrisStrawNinetyFour' has not been observed under all possible environmental conditions. The botanical description of 'DrisStrawNinetyFour' was taken from plants that were five months old. The indicated values represent averages calculated from measurements of several plants. Color references are primarily to The R.H.S. Colour Chart of The Royal Horticultural Society of London (R.H.S.) (2015 edition). Descriptive terminology follows the *Plant Identification Terminology, An Illustrated Glossary*, 2<sup>nd</sup> edition by James G. Harris and Melinda Woolf Harris, unless where otherwise defined.

Classification:

*Species*.—*Fragaria* x *ananassa*.

*Common name*.—Strawberry.

*Denomination*.—'DrisStrawNinetyFour'.

Parentage:

*Female parent*.—'DrisStrawSixtyNine' (U.S. Plant Pat. No. 32,801).

*Male parent*.—Proprietary strawberry plant '575AC159' (unpatented).

Plant:

*Height*.—19.7 cm.

*Diameter*.—26.1 cm.

*Height/width ratio*.—0.75.

*Number of crowns per plant*.—5.

*Growth habit*.—Semi-upright.

*Density of foliage*.—Dense.

*Vigor*.—Strong.

Stolon:

*Diameter at bract*.—3.17 mm.

*Overall color*.—RHS 144C (Strong yellow green).

*Anthocyanin coloration*.—Medium.

*Anthocyanin color (when present)*.—RHS 58A (Moderate purplish red).

*Density of pubescence*.—Sparse.

Fruiting truss:

*Length (from crown to base of terminal flower or fruit)*.—22.5 cm.

*Diameter (at base)*.—0.47 cm.

*Number of berries per fruiting truss*.—4.8.

*Attitude at first picking*.—Semi-erect.



## 3

*Color (at base of truss).*—RHS 144C (Strong yellow green).

Leaf:

*Number of leaflets.*—Three only.

*Color of leaf upper surface.*—RHS NN137 (Greyish olive green). 5

*Color of leaf lower surface.*—RHS 147D (Moderate yellow green).

*Blistering.*—Medium.

*Glossiness.*—Medium. 10

*Variegation.*—Present.

*Terminal leaflet.*—Length: 8.05 cm. Width: 8.4 cm. Length/width ratio: 0.96. Number of teeth per terminal leaflet: 23. Overall shape: Orbicular. Shape of base: Obtuse. Shape of apex: Rounded. Margin: Serrate to crenate. Margin profile: Revolute. Shape in cross section: Concave. 15

*Petiole.*—Length: 16.4 cm. Diameter: 3.6 mm. Overall color: RHS 144C (Strong yellow green). Pubescence: Absent or very sparse. Attitude of hairs: Horizontal. Bract frequency (number present on each petiole): 2. 20

*Petiolule.*—Length: 13.5 mm. Diameter: 2.0 mm. Color: RHS 144C (Strong yellowish green).

*Stipule.*—Length: 35.1 mm. Width: 24.0 mm. Stipule color: RHS 141 (Strong yellow green). Anthocyanin coloration: Absent or very weak. Pubescence: Dense. 25

Inflorescence:

*Number of flowers per plant.*—8.

*Position of inflorescence in relation to foliage.*—Above foliage. 30

*Pedice.*—Attitude of hairs: Horizontal.

*Flower.*—Flower diameter (petal tip to petal tip on non-flattened flower): 20.1 mm. Arrangement of petals: Overlapping. Size of calyx in relation to corolla: Larger. Stamen: Present. Receptacle color: RHS 9A (Vivid yellow). Anther color: RHS N167A (Brownish orange). 35

*Petal.*—Length: 9.9 mm. Width: 11.1 mm. Length/width ratio: 0.89. Number of petals per flower: 6. Color of upper surface: RHS 155C (Greenish white). Color of lower surface: RHS 155C (Greenish white). Overall shape: Orbicular. Shape of apex: Rounded. Margin: Entire. Shape of base: Concavo-convex. 40

*Calyx.*—Diameter (sepal tip to sepal tip, measured on back of flower): 24.7 mm. 45

*Sepal.*—Length: 8.7 mm. Width: 5.4 mm. Number of sepals per flower: 11. Overall shape: Elliptical. Shape of apex: Acuminate. Margin: Entire.

*Flowering.*—Flowering interval: July to December. 50

Fruit:

*Fruit size.*—Length: 55.3 mm. Width: 48.5 mm. Length/width ratio: 1.14.

*Fruit hollow.*—Length: 28.3 mm. Width: 14.2 mm. Length/width ratio: 1.99. 55

*Shape.*—Cordate.

*Difference in shape of terminal and other fruits.*—Moderate.

*Fruit color.*—RHS 34A (Vivid reddish orange).

*Evenness of color.*—Even or very slightly uneven. 60

*Glossiness.*—Medium.

*Evenness of surface.*—Even or very slightly uneven.

*Width of band without achenes.*—Absent or very narrow.

*Position of achenes.*—Level with surface.

*Position of calyx attachment.*—Inserted.

*Attitude of sepals.*—Upwards.

## 4

*Diameter of calyx in relation to diameter of fruit.*—Slightly smaller.

*Adherence of calyx.*—Strong.

*Firmness.*—Firm.

*Color of flesh (excluding core).*—RHS 35B (Moderate reddish orange).

*Evenness of flesh color.*—Slightly uneven.

*Distribution of flesh color.*—Marginal and central.

*Color of core.*—RHS 37C (Moderate yellowish pink).

*Sweetness/soluble solids (in ° brix).*—8.4.

*Titrate acidity (% as citric acid).*—0.69%.

*Individual fruit weight.*—32.4 g/fruit.

*Achenes.*—Number of achenes per fruit: 221. Weight: 0.0008 g/achene. Color of upper (sunward) side: RHS N34 (Moderate red). Color of lower (shaded) side: RHS 144A (Strong yellow green).

*Fruiting.*—Harvest interval: October to December. Type of bearing: Fully remontant. Productivity: 31,202 kg to 48,921 kg of fruit per hectare per season from five-month-old plants when grown in Ventura County, Calif.

Resistance to abiotic stress, pests, and diseases:

*Two-spotted spider mite (tetranychus urticae).*—Moderately resistant.

*Botrytis fruit rot (botrytis cinerea).*—Moderately susceptible.

*Powdery mildew (podosphaera macularis).*—Moderately susceptible.

*Anthracnose crown rot (colletotrichum acutatum).*—Moderately susceptible.

COMPARISON WITH PARENTAL AND  
REFERENCE VARIETIES

‘DrisStrawNinetyFour’ differs from the female parent and reference variety ‘DrisStrawSixtyNine’ (U.S. Plant Pat. No. 32,801) in that ‘DrisStrawNinetyFour’ has medium stolon anthocyanin coloration, absent or very weak stipule anthocyanin coloration, inserted position of calyx attachment on fruit, horizontal attitude of pedicel hairs, and leaf variegation is present, whereas ‘DrisStrawSixtyNine’ has absent or very weak stolon anthocyanin coloration, medium stipule anthocyanin coloration, raised position of calyx attachment on fruit, upwards attitude of pedicel hairs, and leaf variegation is absent. 40

‘DrisStrawNinetyFour’ differs from the male parent proprietary strawberry plant ‘575AC159’ (unpatented) in that ‘DrisStrawNinetyFour’ has a higher yield potential and larger fruit size than ‘575AC159’, as well as improved flavor. 50

‘DrisStrawNinetyFour’ differs from the reference variety ‘DrisStrawTwentyOne’ (U.S. Plant Pat. No. 23,506) in that ‘DrisStrawNinetyFour’ has medium stolon anthocyanin coloration, a moderate difference in the shape of terminal and other fruits, a slightly smaller calyx in relation to the diameter of fruit, horizontal attitude of pedicel hairs, and leaf variegation is present, whereas ‘DrisStrawTwentyOne’ has strong stolon anthocyanin coloration, the difference in shape of terminal and other fruits is none or very slight, has a much larger calyx in relation to the diameter of its fruit, upwards attitude of pedicel hairs, and leaf variegation is absent. 60

We claim:

1. A new and distinct variety of strawberry plant named ‘DrisStrawNinetyFour’ as shown and described herein. 65

\* \* \* \* \*



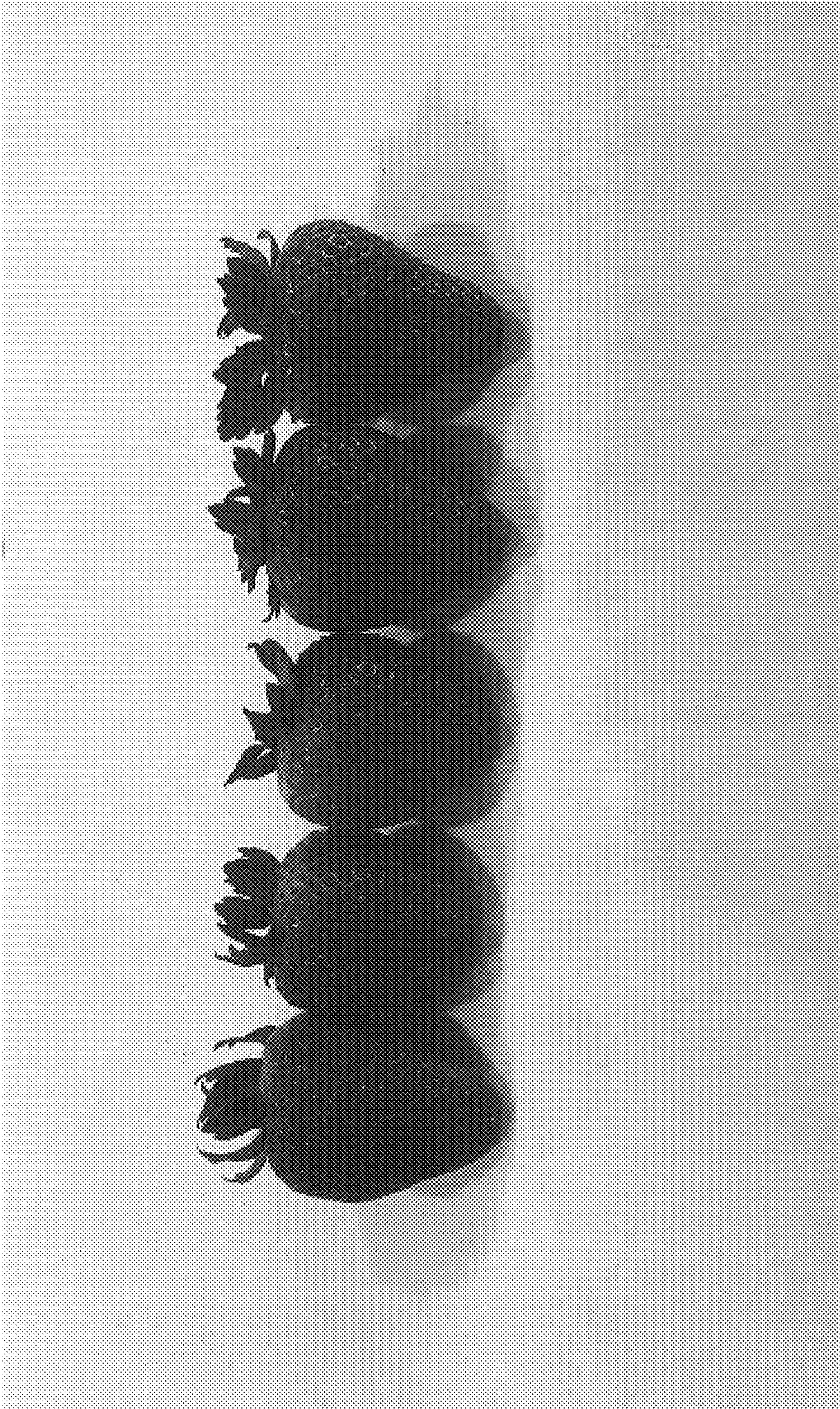


FIG. 1



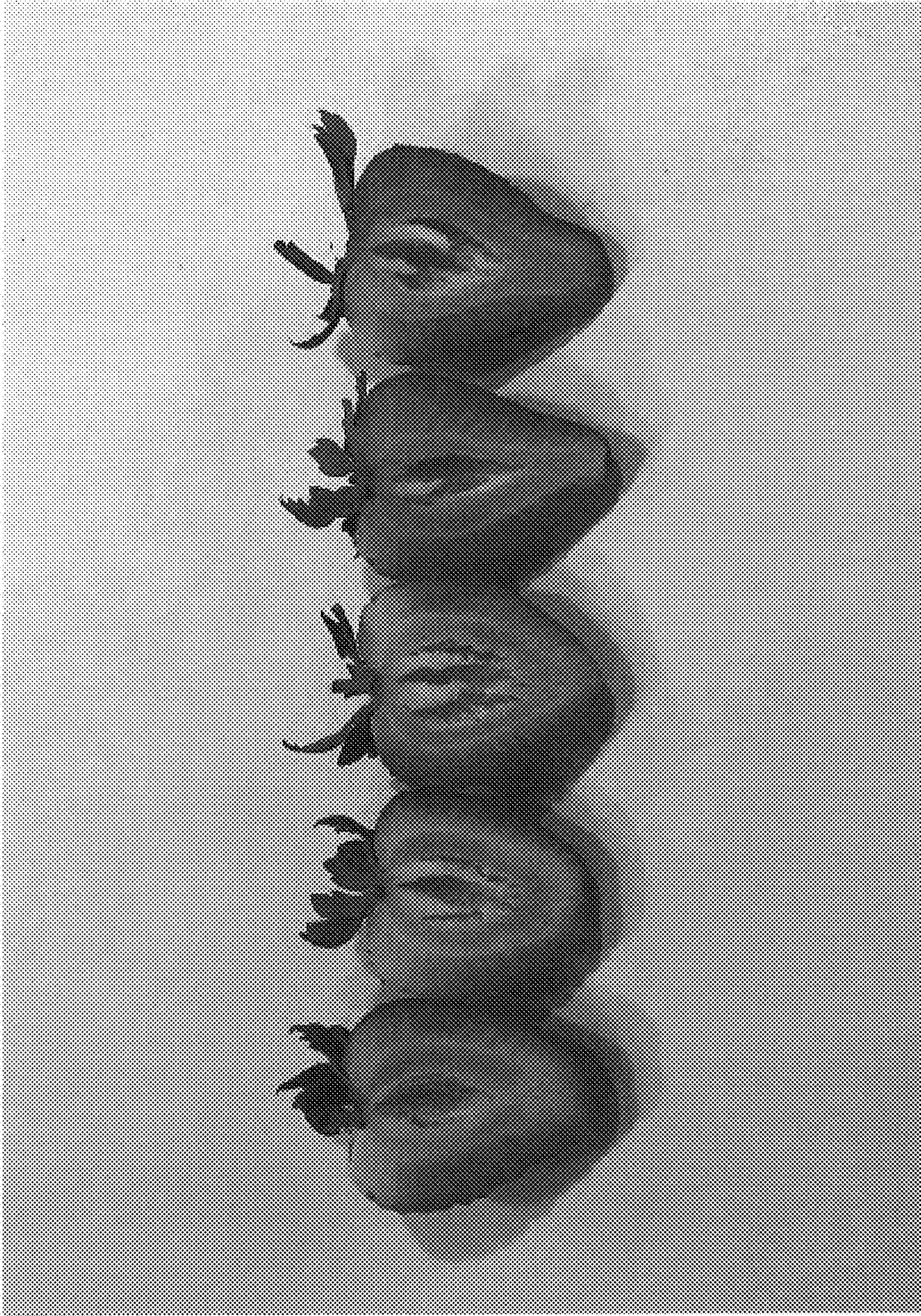


FIG. 2



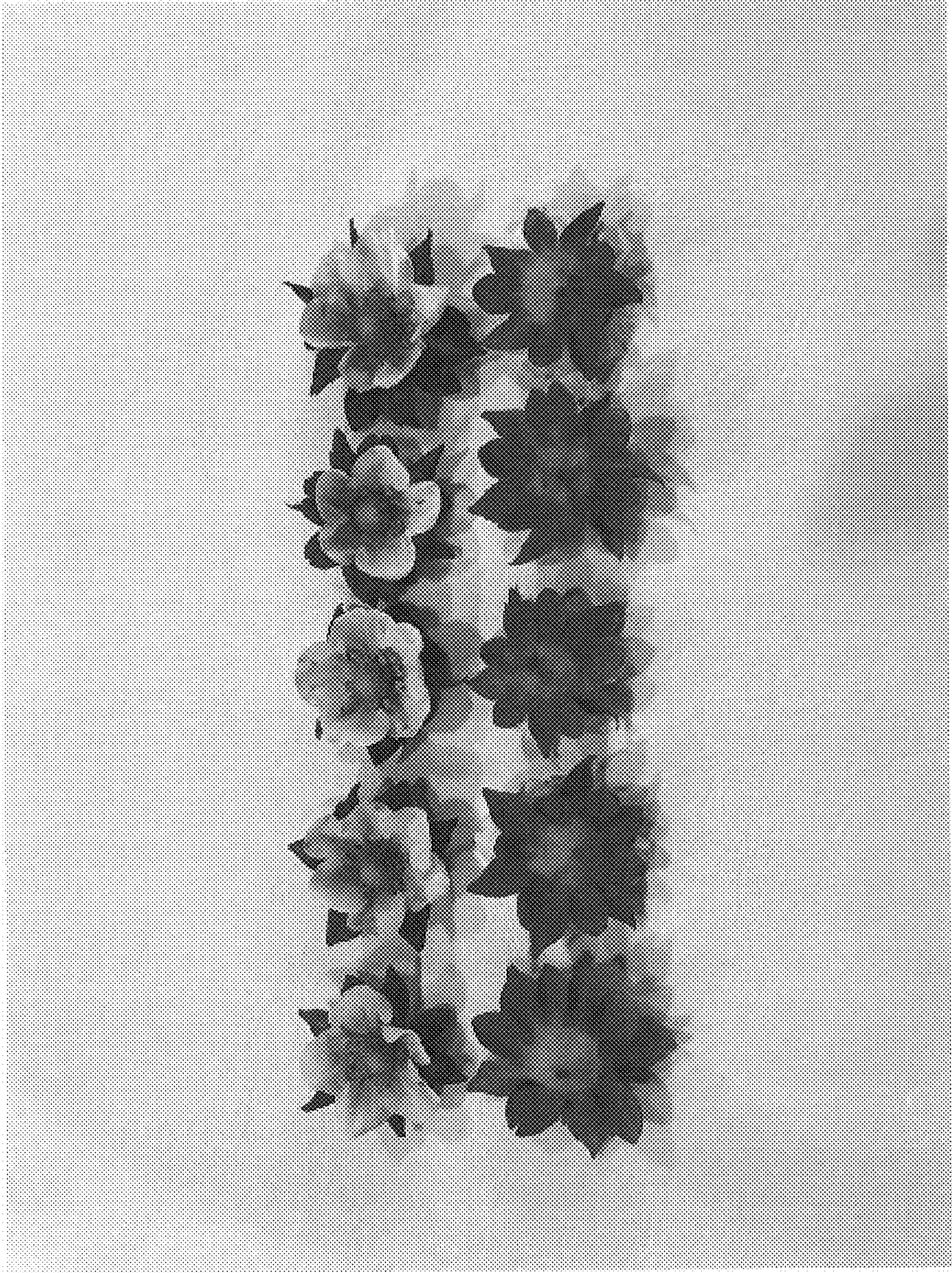


FIG. 3



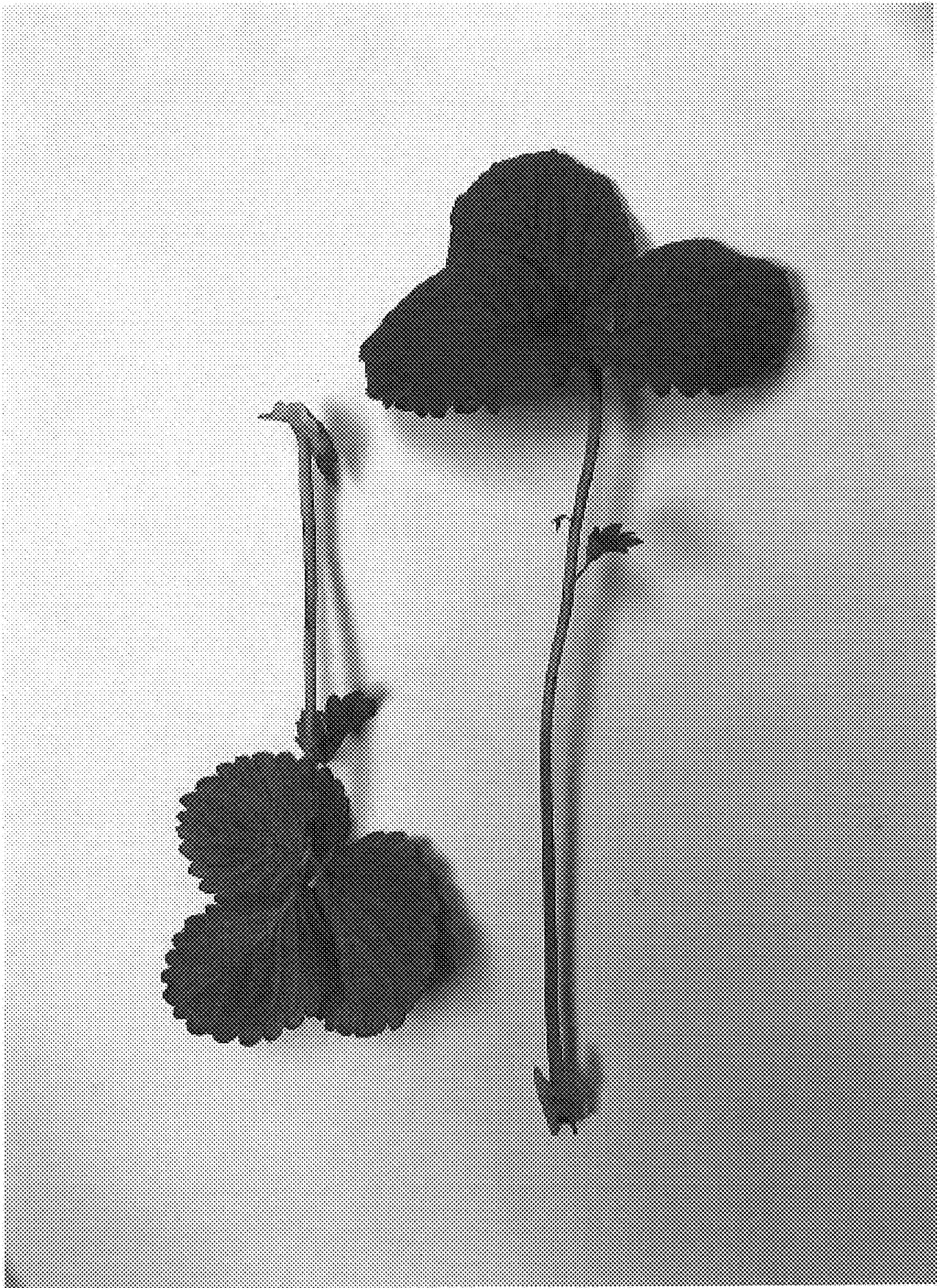


FIG. 4





FIG. 5